

Fig. 4: PSoC based board for toy applications

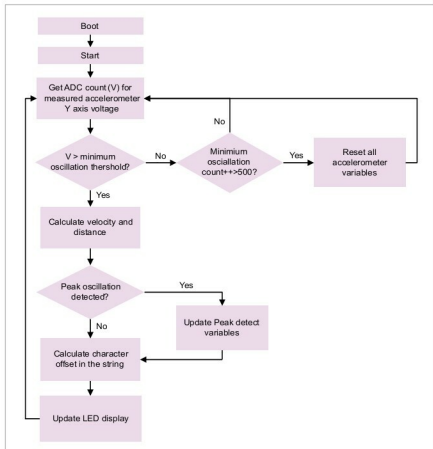


Fig. 5: PSoCRocks application flowchart

tal conversions and commonly-used fixed-function peripherals. The PSoC family is made up of 8-bit (PSoC 1 and PSoC 3) and 32-bit (PSoC4 and PSoC5) MCUs. These

have flash memories up to 256kB, SRAM up to 64kB and internal EEPROM up to 2kB (flash can be used to emulate EEPROM). A PSoC works on ultra-low power mode.

In low-power mode, it consumes less than 1µA current, making it useful when operating a toy in low-power mode and with coin-cell battery input.

A PSoC has an internal RTC component for real-time measurements. It does not require an external clock/oscillator circuitry and supports USB 2.0 interface, allowing the user to interface external memory through USB 2.0. It also supports secure digital card interface.

A PSoC has internal PWM, which is used for controlling and varying the tone of the buzzer for every key press in electronic toy applications. Duty cycle of PWMs is varied based on the speed required as set by the user through the GUI/keypad interface. A PSoC has an internal 8-bit DAC, which can be used for controlling speaker volume, and it also performs mute function for the speaker.

A PSoC has internal op-amps, comparators and 12-bit 1MSPS successive-approximation ADC with differential and single-ended modes including sample-and-hold capability, which is used to measure different sensor inputs, and battery monitoring. So it does not require external amplifiers, ADC and comparators for electronic toy applications.

A PSoC has an operating range of 1.71V to 5.5V, so it can be easily interfaced with external peripherals for other applications. It supports a low operating voltage up to 1.71V and ultra-low power (consumes less than 1µA current) operation along with hibernate and deep-sleep modes, which allow wake-up time versus power trade-off, which, in turn, help support extended battery life/power supply life.

A PSoC supports two independent run-time reconfigurable serial communication blocks with re-configurable I2C, SPI or UART functionality, which are helpful